CV Mid Sems Syllabus

Tuesday | 24 February 2026 | 4:22 PM

- Status : #doing
- Tags : acads | computer vision

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1. Fundamentals

- [ ] What is an Image?
- [ ] What is a Pixel?
- [ ] How an Image is Formed
- [ ] Pin-hole Camera
- [ ] Properties of Camera
- [ ] Focal Length
- [ ] Gaussian Lens Law
- [ ] Object Magnification
- [ ] Depth of Focus
- [ ] Two-Lens Camera
- [ ] Vanishing Point

2. Image Enhancement

A. Spatial Domain

- [ ] Negative Transformation
- [ ] Identity Transformation
- [ ] Log Transformation
- [ ] Power Function Transformation
- [ ] Gamma Correction
- [ ] Thresholding
- [ ] Piecewise Thresholding

- [ ] Image Binarization
- [ ] Bit Plane Slicing
- [ ] Histogram Equalization

B. Frequency Domain (Optional)

- [ ] Discrete Fourier Transform (DFT)
- [ ] 1D DFT
- [ ] 2D DFT
- [ ] Inverse DFT

3. Image Segmentation

- [ ] Region Growing – 4-connected method
- [ ] Region Growing – 8-connected method
- [ ] K-means Clustering

4. Edge Detection (Optional)

- [ ] Sobel Operator
- [ ] Prewitt Operator
- [ ] Canny Edge Detection